Stewart, Calculus: Early Transcendentals

Section 15.9, Exercise 43: drawing a silo

Problem

Use a graphing device to draw a silo consisting of a cylinder with radius 3 and height 10 surmounted by a hemisphere.

Symbols used in this solution

- C is the cylindrical body, D is the hemispherical roof, and $E = C \cup D$ is the whole silo.
- r, θ, z are cylindrical coordinates, with $r = \sqrt{x^2 + y^2}$ the distance from the z-axis. The silo stands on the xy-plane and its axis is the z-axis.
- ρ, θ, φ are spherical coordinates. For the roof it is convenient to measure them from the centre $(0, 0, 10)$ of the sphere: ρ is the distance from that point and φ is the angle from the upward vertical.
- dV is the volume element: $dV = r\,dz\,dr\,d\theta$ in cylindrical coordinates and $dV = \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta$ in spherical coordinates.
- V is the volume, S the total surface area, $h = 10$ the height of the cylindrical part and $R = 3$ the common radius.

Part (a): the drawing, the volume and the surface area

Step 1. The solid

The body is the cylinder

$$C = \{(x, y, z) : x^2 + y^2 \le 9,\ 0 \le z \le 10\}, \tag{1}$$

that is $0 \le r \le 3$, $0 \le \theta \le 2\pi$, $0 \le z \le 10$. The roof is the upper half of the sphere of radius 3 centred at $(0, 0, 10)$:

$$D = \{(x, y, z) : x^2 + y^2 + (z - 10)^2 \le 9,\ z \ge 10\}. \tag{2}$$

In spherical coordinates measured from the centre $(0, 0, 10)$ the roof is simply $\rho \le 3$ with $0 \le \varphi \le \pi/2$ and $0 \le \theta \le 2\pi$. The two parts meet along the circle

$$x^2 + y^2 = 9, \qquad z = 10, \tag{3}$$

which is the rim of the roof. On that circle the wall is vertical and the roof is also vertical: the outward normal of the sphere at the rim is horizontal. The surface is therefore smooth across the rim, so the silo is one closed solid and not two pieces sitting on each other.

The dimensions fix the shape. The diameter is $2R = 6$, the height is $h + R = 10 + 3 = 13$, and the ratio height to diameter is $13/6 \approx 2.17$. A correct drawing must show this proportion; Figure 1 does, because its 3D box is scaled to the true ranges of x, y and z.

Step 2. The volume

The two parts share only a surface of zero volume, so the volumes add. For the body, the inner integration runs along a vertical line of length h, and the outer two integrations run over the disc of radius R in polar form:

$$V_C = \int_0^{2\pi} \int_0^{3} \int_0^{10} r \, dz \, dr \, d\theta = 2\pi \cdot \frac{3^2}{2} \cdot 10 = 90\pi. \tag{4}$$

For the roof, use spherical coordinates centred at $(0, 0, 10)$. The radial limits are $0 \le \rho \le 3$, the angle runs over the upper half, $0 \le \varphi \le \pi/2$, and the azimuth over a full turn:

$$V_D = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{3} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta = 2\pi \cdot \Big[-\cos\varphi\Big]_0^{\pi/2} \cdot \frac{27}{3} = 2\pi \cdot 1 \cdot 9 = 18\pi. \tag{5}$$

Here $\int_0^{\pi/2} \sin\varphi \, d\varphi = 1$ is used, so the hemisphere is exactly half of the sphere of volume $\frac{4}{3}\pi \cdot 27 = 36\pi$. Hence

$$V = 90\pi + 18\pi = 108\pi \approx 339.292. \tag{6}$$

The same number follows from slicing perpendicular to the axis. At height z the cross-section is a disc, and

$$A(z) = \begin{cases} 9\pi, & 0 \le z \le 10, \\ \pi\big(9 - (z-10)^2\big), & 10 \le z \le 13, \end{cases} \qquad V = \int_0^{13} A(z) \, dz. \tag{7}$$

The first part gives 90π and the second gives $\pi\big[9u - u^3/3\big]_0^3 = 18\pi$ with $u = z - 10$. The slice view is the reason the roof contributes less than the body although it is just as wide: the disc shrinks to a point at the top.

Step 3. The surface area

Three surfaces form the boundary when the silo is closed at the bottom:

$$\text{lateral wall:} \quad 2\pi Rh = 2\pi \cdot 3 \cdot 10 = 60\pi, \tag{8}$$

$$\text{hemispherical roof:} \quad 2\pi R^2 = 2\pi \cdot 9 = 18\pi, \tag{9}$$

$$\text{floor:} \quad \pi R^2 = 9\pi. \tag{10}$$

The roof area is one half of the area $4\pi R^2$ of the full sphere. Therefore

$$S = 60\pi + 18\pi + 9\pi = 87\pi \approx 273.319 \tag{11}$$

for a closed silo, and $S = 78\pi \approx 245.044$ if the floor is left out. Both conventions are in use; the closed value is quoted in the box below.

Step 4. Check: Pappus and Monte Carlo

The meridian half-section of the silo (Figure 2) is the rectangle $0 \le x \le 3$, $0 \le z \le 10$ together with the quarter disc of radius 3 centred at $(0, 10)$. Its area is $30 + \frac{9}{4}\pi$ and its first moment about the axis is

$$30 \cdot \frac{3}{2} + \frac{9\pi}{4} \cdot \frac{4 \cdot 3}{3\pi} = 45 + 9 = 54. \tag{12}$$

Pappus's theorem then gives, without any integration,

$$V = 2\pi \cdot 54 = 108\pi, \qquad S = 2\pi\big(\underbrace{10 \cdot 3}_{\text{wall}} + \underbrace{3 \cdot \tfrac{3}{2}}_{\text{floor}} + \underbrace{\tfrac{3\pi}{2} \cdot \tfrac{6}{\pi}}_{\text{roof}}\big) = 2\pi(30 + 4.5 + 9) = 87\pi. \tag{13}$$

Table 1 collects the numeric checks. A Monte Carlo estimate inside the bounding cylinder of radius 3 and height 13 uses the fact that a point with $z \le 10$ is always inside the silo, and a point with $z > 10$ is inside when $r^2 + (z - 10)^2 \le 9$.

Table 1: Checks of the volume and of the surface area of the silo.

Quantity	Method	Value
V	exact, `sympy`	$108\pi = 339.292006588$
V	slice integral $\int_0^{13} A(z)\,dz$	339.292006588
V	Pappus, $2\pi \cdot 54$	339.292006588
V	Monte Carlo, 4×10^6 points	339.2865 ± 0.0490
V	fraction of the bounding cylinder	$12/13 = 0.9231$
S	exact, `sympy`	$87\pi = 273.318560862$
S	Pappus, $2\pi(30 + 4.5 + 9)$	273.318560862

$$\boxed{V = 108\pi \approx 339.29, \qquad S = 87\pi \approx 273.32} \tag{14}$$

Step 5. The drawing

Figure 1 is the perspective drawing and Figure 2 is the meridian cross-section with the dimensions. The drawing is made from parametric formulas, so every plotted point satisfies the equation of its surface:

$$\text{wall:} \quad x = 3\cos\theta, \quad y = 3\sin\theta, \quad 0 \le z \le 10, \tag{15}$$

$$\text{roof:} \quad x = 3\sin\varphi\cos\theta, \quad y = 3\sin\varphi\sin\theta, \quad z = 10 + 3\cos\varphi, \quad 0 \le \varphi \le \tfrac{\pi}{2}, \tag{16}$$

$$\text{floor:} \quad x = s\cos\theta, \quad y = s\sin\theta, \quad z = 0, \quad 0 \le s \le 3. \tag{17}$$

The rim circles at $z = 0$ and $z = 10$ are drawn separately, because they are the two visible edges of the body and they make the roundness of the solid clear.

How to solve problems of this kind

1. Write down each surface in its simplest coordinate system. A vertical cylinder is $r = R$; a sphere centred at $(0, 0, 10)$ is $\rho = R$ in spherical coordinates with that centre.

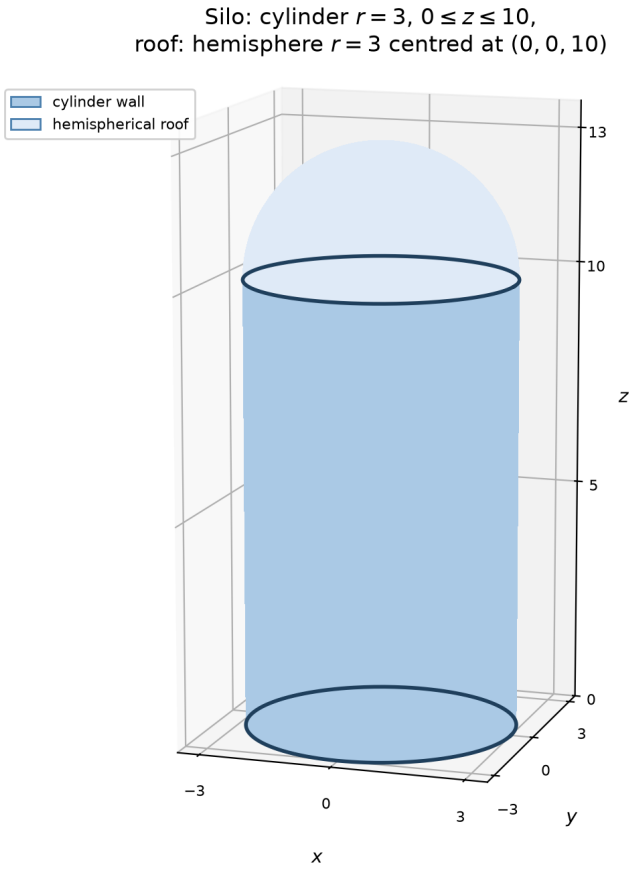


Figure 1: The silo drawn to scale: a cylinder of radius 3 and height 10 with a hemispherical roof of radius 3 centred at $(0, 0, 10)$. The 3D box is scaled to the data ranges, so the height-to-diameter ratio $13 : 6$ is correct. The heavy circles are the rim at $z = 0$ and at $z = 10$.

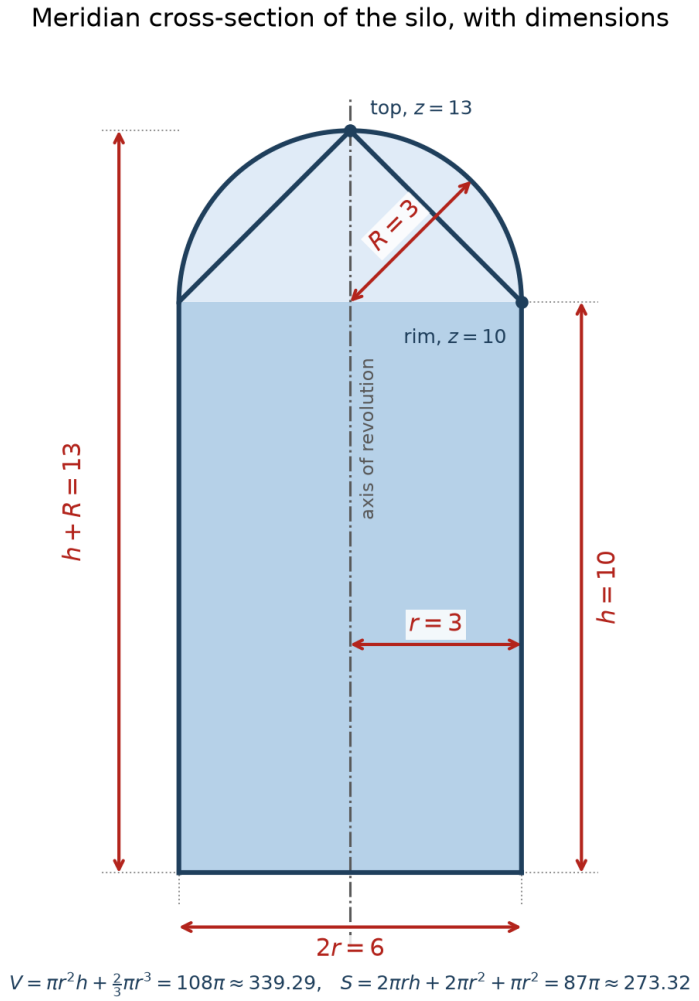


Figure 2: Meridian cross-section of the silo. The body is the rectangle of width $2r = 6$ and height $h = 10$; the roof is the semicircle of radius $R = 3$ centred at the top of the axis. Rotating this section about the axis gives the silo. The dimensions $r = 3$, $h = 10$, $R = 3$ and $h + R = 13$ are marked. The exact volume and surface area are quoted at the bottom of the figure.

2. Find where the surfaces meet. Here the meeting set is the circle $r = 3$, $z = 10$. Check that the tangents agree there, so that the solid really has one closed boundary.

3. Draw the outline first and the shading second. For a surface of revolution, draw the meridian section, then sweep it. Scale the 3D box to the true ranges of the coordinates, otherwise the drawing lies about the proportions.

4. For the volume, add the volumes of the pieces. Use the coordinate system that makes each piece simple: cylinders for the body, spherical coordinates for the roof.

5. Check the volume a second way: slice perpendicular to the axis, or use Pappus's theorem on the meridian section, or run a Monte Carlo estimate.

6. For the surface area, count every piece of the boundary and say whether the flat bottom is included. A hemisphere has half the area of a sphere.